

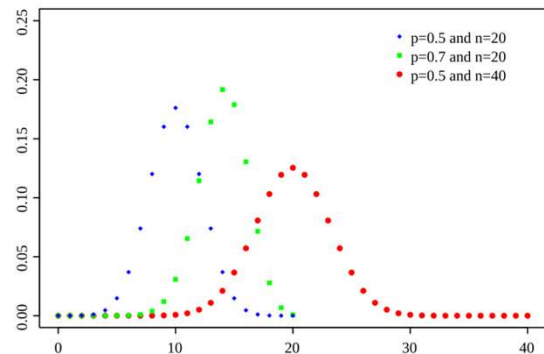
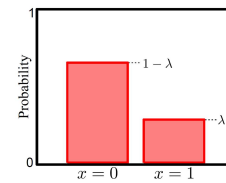
Agenda

○ Last time

- Some distributions used over and over again
 - If we know the mean and variance, we can understand these better
- Bernoulli

○ Today

- Binomial
- Geometric
- Hypergeometric
- Moment generation functions



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Bernoulli

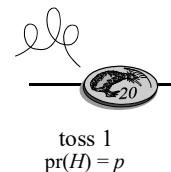
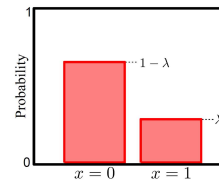
○ Some distributions used over and over again

○ Last time

- $W \sim \text{Bern}(\lambda)$
 - Expectation: λ , Variance: $\lambda(1 - \lambda)$

○ Physical analogy

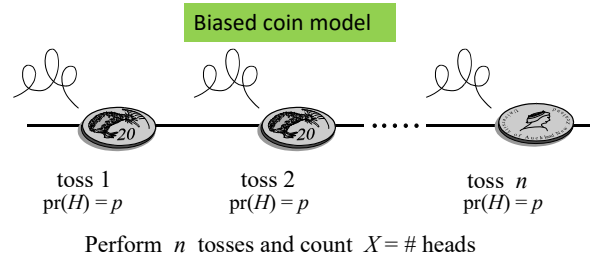
- Biased coin



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Binomial

- Physical model for situations of a series of trials
 - Each trial has only two outcomes: success and failure;
 - $p = \text{pr}(\text{success})$ is the same for every trial; and
 - trials are independent
- Expectation and Variance?



$$P(X = i) = \binom{n}{i} p^i (1 - p)^{n-i}$$

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Example: US Presidents

- The Economist magazine reported that US presidents had 61 daughters and ~90 sons
- Expected value is approximately 75
- What is the probability that there are 89 or more boys?
 - $(1 - \text{binocdf}(89, 150, 0.5))$ in octave
- Probability theory calculates that the probability of having 89 or more sons in 151 'experiments' is 0.008799 (~0.8%), a very small number

Were US presidents special?

Overview of Presidential Children

- Total Recorded (Historical baseline up to Clinton): 89 boys and 61 girls (150 total)

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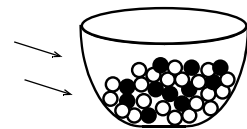
Hypergeometric

- Urn model is actually “hypergeometric” distribution
- We chose Y balls *without replacement* and ask how many are black
- However, the binomial approximation to the urn Model Approximation is very useful
 - Y items sampled, count X as the number of people which have a characteristic
 - Approximation: $X \sim \text{Bin}(Y, M/N)$

N balls in an urn, of which there are

M black balls

$N - M$ white balls



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Example: Doctors and Exercise

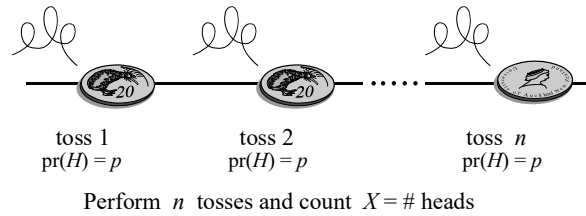
- A 2018 study followed a nationwide cohort of Norwegian doctors. In its 2014 survey, 307 of 526 respondents reported that they asked patients about exercise habits
 - You are choosing a doctor for your pre-diabetic uncle
 - You have a choice of three doctors in your clinic
 - What is the probability that none of the three doctors will ask about exercise?
- This problem fails the binomial requirement of “sampling with replacement” and independence of opinion
 - But we will use the binomial approximation since $3/526$ is very small.
 - $P(X \leq 0) = \text{binocdf}(0, 3, q=219/526) = 19.8\%$

Is the clinic a good one?

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Binomial distribution

- Expectation
- Variance



$$P(X = i) = \binom{n}{i} p^i (1 - p)^{n-i}$$

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Binomial Distribution

- Defining indicator variables, a binomial random variable X is the *sum* of indicator random

$$P(X = i) = \binom{n}{i} p^i (1 - p)^{n-i}$$

$$X_i = \begin{cases} 1, & \text{trial } i \text{ succeeds (probability } p) \\ 0, & \text{otherwise} \end{cases}$$

$$E(X) = E\left(\sum_{i=1}^n X_i\right) = np$$

$$\text{Var}(X) = \text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Var}(X_i) = np(1 - p)$$

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Geometric Distribution

$$P_X(k) = \begin{cases} p(1-p)^{k-1} & \text{for } k = 1, 2, 3, \dots \\ 0 & \text{otherwise} \end{cases}$$

- $X \sim \text{Geom}(p)$
- A sequence of independent Bernoulli trials until you observe the first success
- “Memoryless”
- $E[X] = ?$
- $E[X] = 1.p + (1-p)[1 + E[X]]$
- $\text{Var}(X) = (1-p)/p^2$

